

Steinbock

A two-fabric wall quilt of the alpine ibex under the stars of Capricornus

Steinbock is German for the alpine ibex and for the sign of Capricorn. The quilt speaks the same language as the patchwork lion: two fabrics, and nothing but squares, rectangles and half-square triangles on a 2" grid. The ibex's horn is a staircase of triangles with three ridges, the beard is a single triangle and the eye is one dark square. Above the ibex, the four brightest stars of the constellation stand in their true order.

FINISHED SIZE

40 × 40"

GRID

20 × 20 units of 2"

PIECES

69 squares and rectangles,
45 half-square triangles

FABRICS

2, plus backing

LEVEL

Confident beginner

TECHNIQUES

Rotary cutting, half-square
triangles, straight-seam
piecing



A star chart in the top band

Read the band as a star chart: you are facing south on an autumn evening, when Capricornus sits low over the horizon, so east is on the left and west is on the right.

Deneb Algedi, the brightest star of the constellation, gets the big sawtooth star. Its name is Arabic for “the tail of the goat”, so it shines above the ibex’s tail. **Nashira** sits next to it. **Dabih** and **Algedi** mark the goat’s head and horns in the old star maps, so their pinwheels hang above the ibex’s head.

Down in the night window, an **edelweiss** star grows under the ibex’s chin. It is a friendship star block, reversed out in light on dark.

The ibex faces right and the lion faces left. Hang the ibex to the left of the lion and the two look at each other.



δ Deneb Algedi

γ Nashira

β Dabih

α Algedi

Capricornus as it appears in the sky (top) and the four stars the band shows (bottom). Dot sizes follow brightness.

MATERIALS

What you need

MATERIAL	AMOUNT	USED FOR
 Fabric L, light	1⅞ yd	ibex, edelweiss, frame, sky band, half of every triangle
 Fabric D, dark	1½ yd	Night window, star centres, half of every triangle, binding
Backing	48 × 48"	Piece it if your fabric is narrower
Batting	48 × 48"	Thin cotton or cotton blend keeps a wall quilt flat
Hanging sleeve	8½ × 38"	Leftover backing works
Thread	Neutral grey cotton, 50 wt	Plus quilting thread to match each fabric

Contrast is everything. Fabric L should be clearly lighter than fabric D. Photograph them side by side in black and white: if they look alike, the ibex will disappear. A mottled or hand-dyed light fabric gives the shimmer of the lion’s gold. A near-solid dark lets the stars read crisply.

Tools: rotary cutter, mat and long ruler, a square ruler of at least 6½" for trimming triangles, a fabric marker, pins, and sticky notes for the piece codes.

How this pattern works

Seam allowance

All sizes include $\frac{1}{4}$ " seams. Test before you start: sew six $2\frac{1}{2}$ " scrap squares into a row. It must be $12\frac{1}{2}$ " long, exactly as long as a D6 strip. Shorter means your seam is too wide, longer means too narrow. This is the check that matters most: $\frac{1}{32}$ " off per seam makes the busiest sections $\frac{3}{4}$ " shorter than the plain strips.

Grid

One grid unit is 2" finished. Every piece is a whole number of units, so every seam lines up with the grid.

Piece codes

L = light fabric, D = dark fabric, numbered from smallest to largest. Label each stack as you cut.

WOF

Width of fabric: cut strips selvage to selvage. The plans need 40" of usable width.

Pressing

Press triangle seams toward the dark fabric. Press every other seam open: open seams keep the many intersections flat.

Diagrams

All diagrams show the right side of the fabric. Gaps show which pieces are joined first: the narrower the gap, the earlier the seam.

CUTTING

Cut the strips, then the pieces

Each diagram is one strip cut across the width of the fabric, with the pieces in cutting order. Grey areas are left over. Pieces marked “trimmed” are cut to length from a wider strip and then trimmed to width.

The four long strips **D10** **L6** **L7** **L8** are cut $\frac{3}{4}$ " longer than they finish (the pale end in the diagrams). Each one is sewn to a pieced part, so you trim it to the length you measure on that part.



Fabric L, light

$1\frac{1}{8}$ yd · ibex, frame, sky band

Cut 2 strips $4\frac{1}{2}$ " × WOF, 2 strips 3" × WOF for the triangle squares, 7 strips $2\frac{1}{2}$ " × WOF.

1 $4\frac{1}{2}$ " strip → L11 at $20\frac{1}{2}$ " · 2 × L10 at $8\frac{1}{2}$ " · L2 at $2\frac{1}{2}$ "

2 $4\frac{1}{2}$ " strip → L9 at $4\frac{1}{2}$ " · L7 at $33\frac{3}{4}$ " ($\frac{3}{4}$ " extra, trim to fit later), trimmed to $2\frac{1}{2}$ " wide

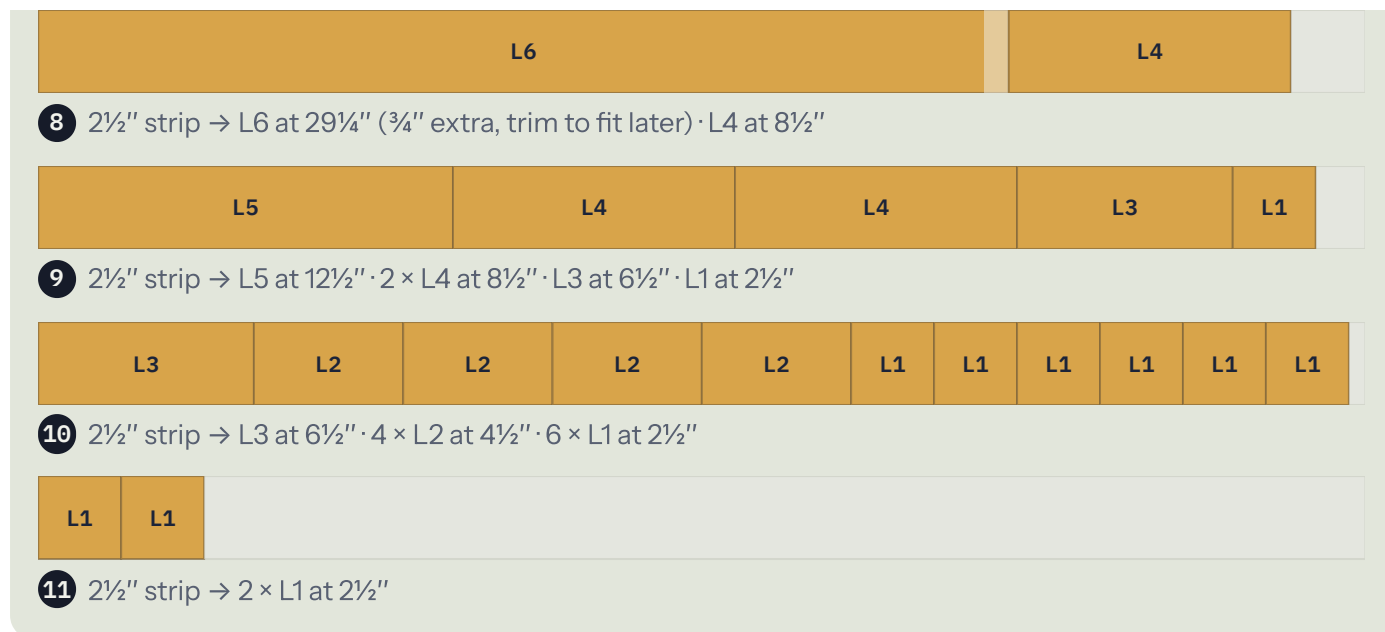
3 3" strip → 13 squares 3 × 3 for the triangles

4 3" strip → 10 squares 3 × 3 for the triangles

5 $2\frac{1}{2}$ " strip → L8 at $37\frac{1}{4}$ " ($\frac{3}{4}$ " extra, trim to fit later) · L1 at $2\frac{1}{2}$ "

6 $2\frac{1}{2}$ " strip → L7 at $33\frac{3}{4}$ " ($\frac{3}{4}$ " extra, trim to fit later) · L3 at $6\frac{1}{2}$ "

7 $2\frac{1}{2}$ " strip → L6 at $29\frac{1}{4}$ " ($\frac{3}{4}$ " extra, trim to fit later) · L4 at $8\frac{1}{2}$ "



Piece list for fabric L

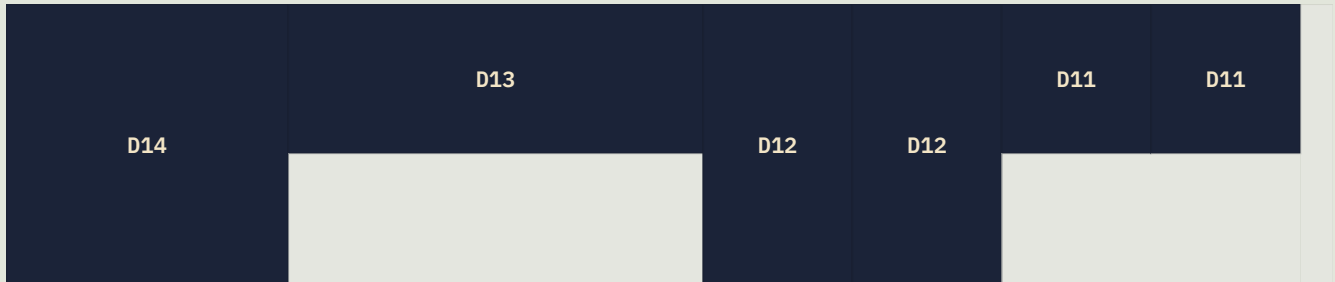
CODE	CUT SIZE	QTY	USED IN
L1	2½" × 2½"	10	Deneb Algedi ×4, section 3 ×2, section 4 ×2, section 6, Edelweiss
L2	2½" × 4½"	5	sky band ×2, section 2, section 5, section 6
L3	2½" × 6½"	3	section 8, section 10 ×2
L4	2½" × 8½"	4	section 11 ×4
L5	2½" × 12½"	1	section 7
L6	2½" × 28½"	2	sky band ×2
L7	2½" × 32½"	2	frame ×2
L8	2½" × 36½"	1	frame
L9	4½" × 4½"	1	sky band
L10	4½" × 8½"	2	sky band ×2
L11	4½" × 20½"	1	section 9



Fabric D, dark

1½ yd · night window, star centres, binding

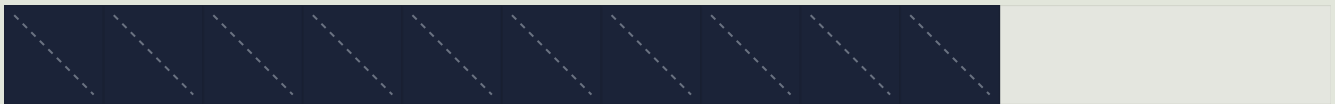
Cut 1 strip 8½" × WOF, 2 strips 3" × WOF for the triangle squares, 8 strips 2½" × WOF, 5 strips 2½" × WOF for the binding.



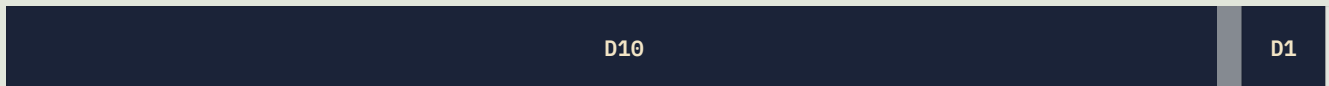
- 1 8½" strip → D14 at 8½" · D13 at 12½", trimmed to 4½" wide · 2 × D12 at 4½" · 2 × D11 at 4½", trimmed to 4½" wide



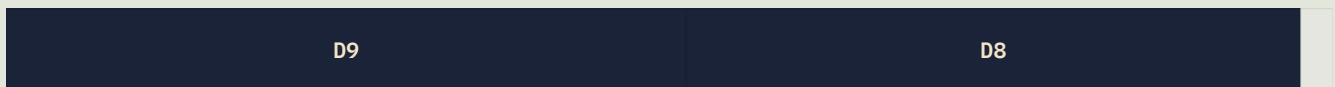
- 2 3" strip → 13 squares 3 × 3 for the triangles



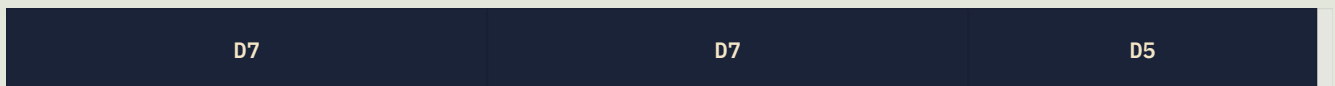
- 3 3" strip → 10 squares 3 × 3 for the triangles



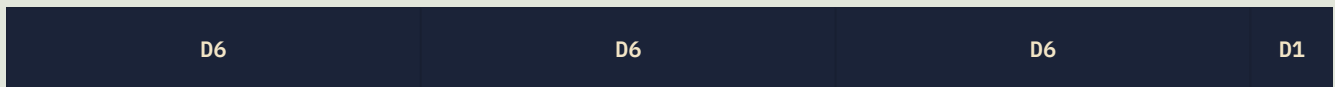
- 4 2½" strip → D10 at 37¼" (¾" extra, trim to fit later) · D1 at 2½"



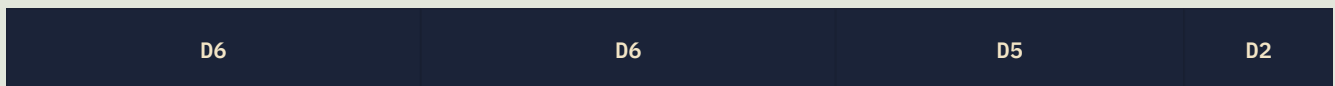
- 5 2½" strip → D9 at 20½" · D8 at 18½"



- 6 2½" strip → 2 × D7 at 14½" · D5 at 10½"



- 7 2½" strip → 3 × D6 at 12½" · D1 at 2½"



- 8 2½" strip → 2 × D6 at 12½" · D5 at 10½" · D2 at 4½"



- 9 2½" strip → 4 × D4 at 8½" · D2 at 4½"

D3	D3	D3	D2	D2	D2	D2	D1
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10 2½" strip → 3 × D3 at 6½" · 4 × D2 at 4½" · D1 at 2½"

D1	D1	D1
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11 2½" strip → 3 × D1 at 2½"

Piece list for fabric D

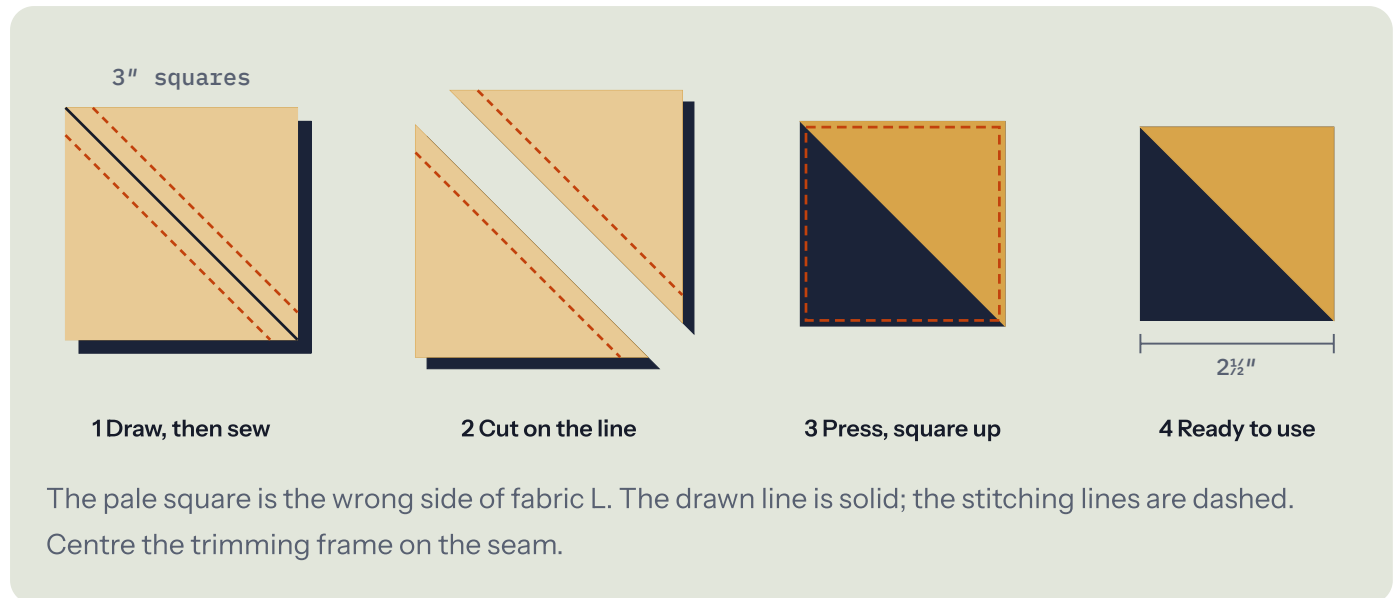
CODE	CUT SIZE	QTY	USED IN
D1	2½" × 2½"	6	section 6, section 8, Edelweiss ×4
D2	2½" × 4½"	6	section 4, section 6, section 7, section 8, section 10 ×2
D3	2½" × 6½"	3	section 5, section 8, section 11
D4	2½" × 8½"	4	section 5, section 11 ×3
D5	2½" × 10½"	2	section 4, section 8
D6	2½" × 12½"	5	section 3 ×2, section 4, section 5, section 10
D7	2½" × 14½"	2	section 2 ×2
D8	2½" × 18½"	1	section 7
D9	2½" × 20½"	1	section 6
D10	2½" × 36½"	1	section 1
D11	4½" × 4½"	2	Deneb Algedi, section 9
D12	4½" × 8½"	2	section 11 ×2
D13	4½" × 12½"	1	section 9
D14	8½" × 8½"	1	section 11

Backing and batting: cut both to 48 × 48". Hanging sleeve: one strip 8½ × 38".

STEP 1

Make 45 half-square triangles

Every triangle unit in the quilt is the same: half light, half dark. Only the way it is turned changes, and every diagram shows it by where the light half sits. The cutting plan gives 23 pairs of squares, so you will make 46 units, with one to spare.



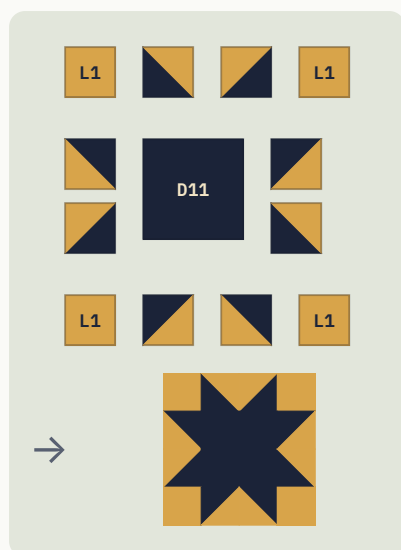
1. Draw a diagonal line on the wrong side of each light 3" square.
2. Place it right sides together on a dark square. Sew 1/4" from the line on both sides.
3. Cut on the line. Press the seam toward the dark fabric.
4. Trim to 2 1/2 x 2 1/2". Lay the 45° line of the square ruler on the seam so the diagonal runs exactly corner to corner.

Faster: eight at a time. Instead of the small squares, cut 6 light and 6 dark squares of 6". Draw both diagonals, sew on both sides of both lines, cut horizontally, vertically and along both diagonals, then trim. That gives 48 units. Their edges are on the bias, so press gently and do not pull.

STEP 2

Piece the stars

Five small blocks: the sawtooth star of Deneb Algedi, three pinwheel stars and the edelweiss. Sew them first. In the later diagrams each block appears as one unit inside a dashed outline.

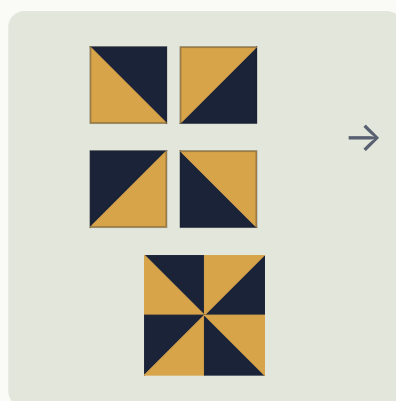


Deneb Algedi

SAWTOOTH STAR • MAKE 1

You need 8 triangles, 4 × **L1**, 1 × **D11**.

1. Join the top row and the bottom row: an L1, two triangles, an L1. The two dark points of each pair meet at the centre seam.
2. Join two triangles top to bottom for each side, then sew them to the left and right of D11.
3. Join the three rows. The block measures $8\frac{1}{2}'' \times 8\frac{1}{2}''$.

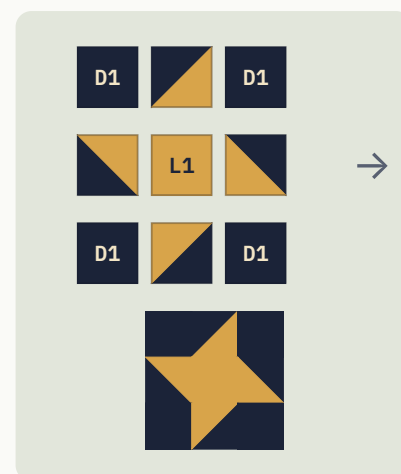


Nashira, Dabih, Algedi

PINWHEELS • MAKE 3

You need 4 triangles per pinwheel.

1. Join the triangles in two rows of two, exactly as shown, so all three pinwheels spin the same way.
2. Join the rows. Pin the centre so the four points meet. Each block measures $4\frac{1}{2}'' \times 4\frac{1}{2}''$.
3. Eight seams meet in the centre. Press the last seam open, or spin it: undo the two stitches inside the seam allowance at the centre and press the seams around in a circle, so the centre lies flat.



Edelweiss

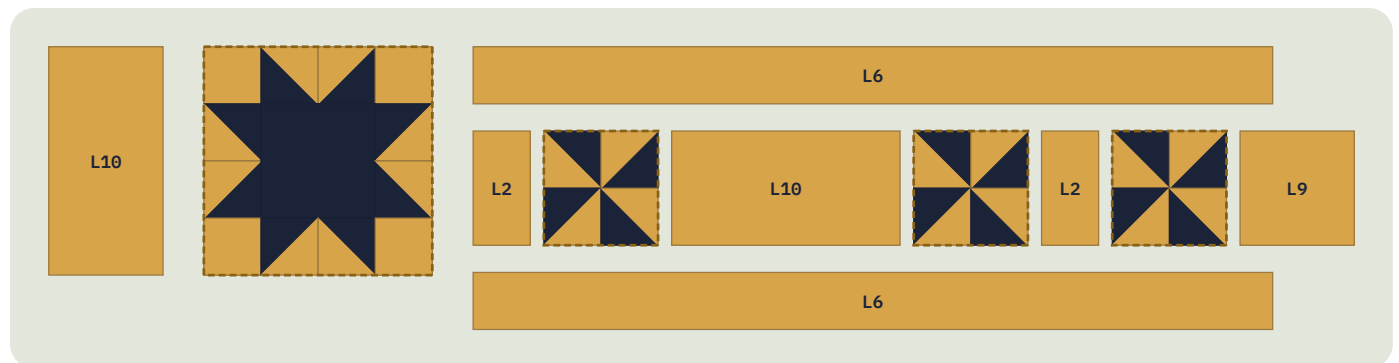
FRIENDSHIP STAR • MAKE 1

You need 4 triangles, 4 × **D1**, 1 × **L1**.

1. Join three rows of three: D1, triangle, D1 / triangle, L1, triangle / D1, triangle, D1.
2. Join the rows. The block measures $6\frac{1}{2}'' \times 6\frac{1}{2}''$.

STEP 3

Assemble the sky band



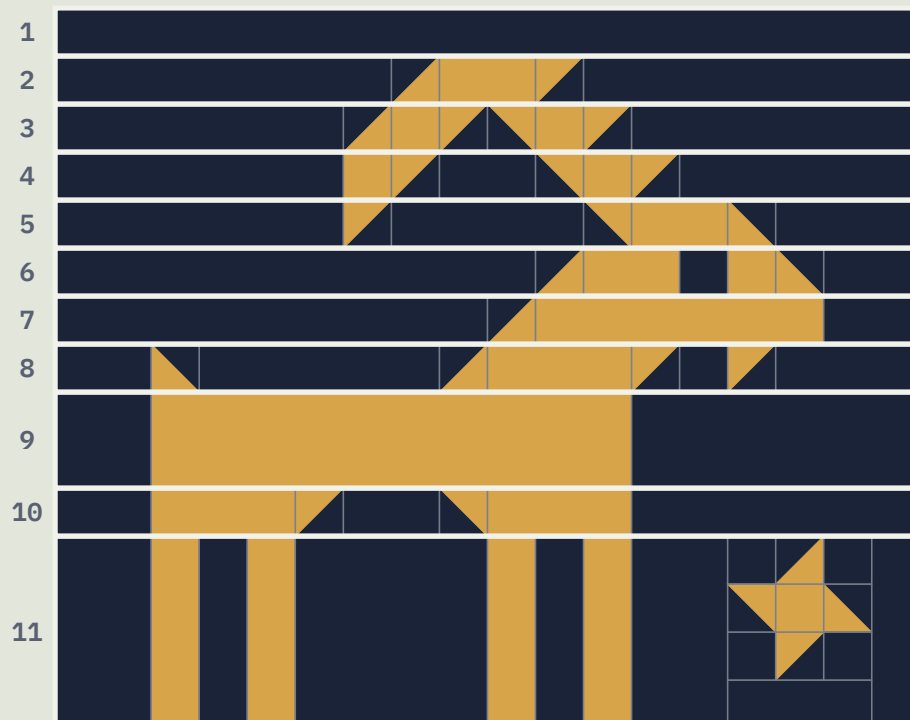
1. Middle row, left to right: **L2** **Nashira** **L10** **Dabih** **L2** **Algedi** **L9**. It measures $28\frac{1}{2}" \times 4\frac{1}{2}"$. Note that L10 lies on its side here.
2. Measure the middle row through its centre and trim both **L6** strips to that length. Sew one to the top and one to the bottom. The unit measures $28\frac{1}{2}" \times 8\frac{1}{2}"$.
3. Join **L10**, the Deneb Algedi block and this unit, left to right.

The finished band measures $40\frac{1}{2}" \times 8\frac{1}{2}"$. Its light background and the light side strips of the frame become one colour field, so the stars float in a light sky.

STEP 4

Piece the night window in 11 sections

The window is pieced as eleven horizontal sections. Sew each section, press the seams open and check its size before going on. Then join sections 1 to 11 from top to bottom.



The window and its sections. Thin lines are the seams inside each section.

In each diagram below, the pieces are spread apart along the seams you sew. Most sections are a single row joined left to right. Section 11 has one column to make first: the edelweiss block over a

D3.

Triangles appear as small icons in the sewing order. For example,  is a triangle unit with the light half at the top right.

Keep the lines straight. The triangle marks ▼ ▲ in each diagram show the 19 seams where the horn, head, neck, body or legs continue into the next section. Pin those first, then both ends, then ease in any difference between the pins. If a section is more than 1/8" off its size, re-sew a seam rather than stretching it.

Sections 10 and 11 have the most pin points. Sew them first as a test: if the legs line up there, the rest will too.

The finished window measures 36½" × 30½".

1

Section 1

Night sky above the horn: a single strip.

Measures $36\frac{1}{2}" \times 2\frac{1}{2}"$



D10

One piece: **D10**. It is cut $\frac{3}{4}"$ long; trim it to the width of section 2 before you join the two.

2

Section 2

The crown of the horn.

Measures $36\frac{1}{2}" \times 2\frac{1}{2}"$



D7

L2

D7

Join left to right **D7**  **L2**  **D7**

Pin points: 1 with section 3 ▲

3

Section 3

The horn arches over. First ridge on the right.

Measures $36\frac{1}{2}" \times 2\frac{1}{2}"$



D6

L1

L1

D6

Join left to right **D6**  **L1**   **L1**  **D6**

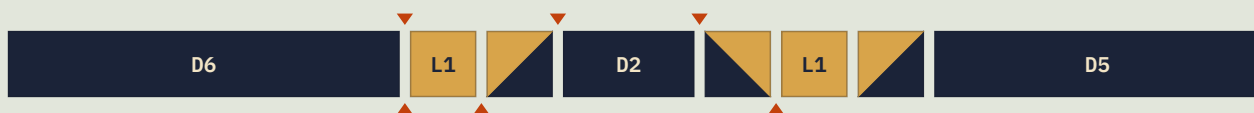
Pin points: 1 with section 2 ▼ · 3 with section 4 ▲

Section 4

4

Horn tip on the left, second ridge on the right.

Measures $36\frac{1}{2}" \times 2\frac{1}{2}"$



Join left to right **D6** **L1** **D2** **L1** **D5**

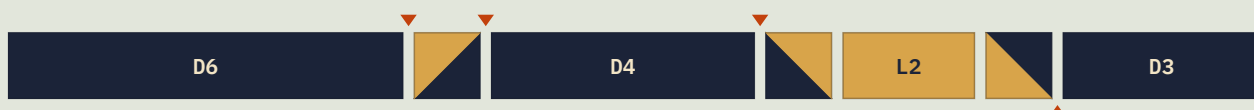
Pin points: 3 with section 3 ▼ · 3 with section 5 ▲

Section 5

5

Horn tip, third ridge and the top of the head.

Measures $36\frac{1}{2}" \times 2\frac{1}{2}"$



Join left to right **D6** **D4** **L2** **D3**

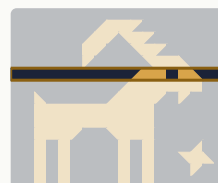
Pin points: 3 with section 4 ▼ · 1 with section 6 ▲

Section 6

6

Back of the head, the eye and the nose.

Measures $36\frac{1}{2}" \times 2\frac{1}{2}"$



Join left to right **D9** **L2** **D1** **L1** **D2**

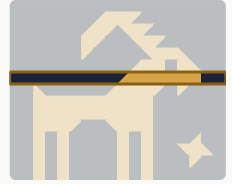
Pin points: 1 with section 5 ▼ · 2 with section 7 ▲

7

Section 7

Neck and muzzle.

Measures $36\frac{1}{2}" \times 2\frac{1}{2}"$



Join left to right **D8**  **L5** **D2**

Pin points: 2 with section 6 ▼ · 1 with section 8 ▲

8

Section 8

Tail, withers, throat and beard.

Measures $36\frac{1}{2}" \times 2\frac{1}{2}"$



Join left to right **D2**  **D5**  **L3**  **D1**  **D3**

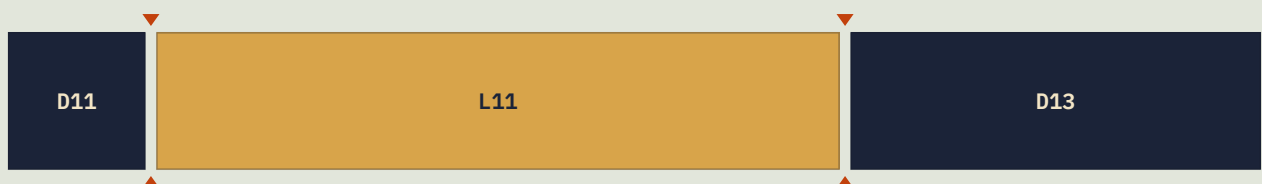
Pin points: 1 with section 7 ▼ · 2 with section 9 ▲

9

Section 9

The body: one long L11.

Measures $36\frac{1}{2}" \times 4\frac{1}{2}"$



Join left to right **D11** **L11** **D13**

Pin points: 2 with section 8 ▼ · 2 with section 10 ▲

10

Section 10

The belly curve.

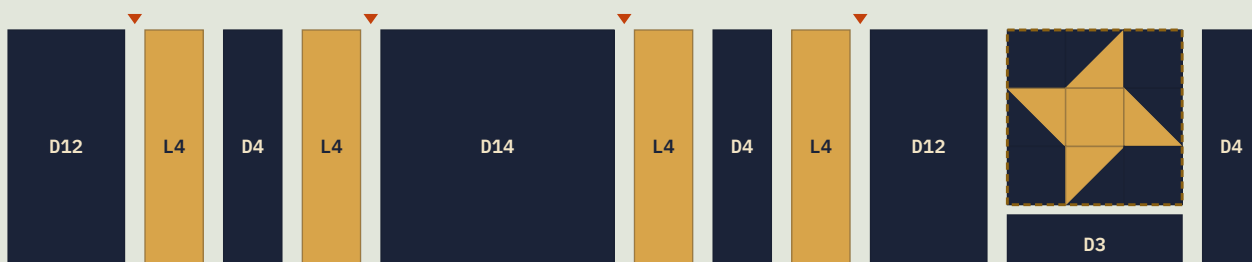
Measures $36\frac{1}{2}'' \times 2\frac{1}{2}''$ Join left to right **D2** **L3**  **D2**  **L3** **D6**

Pin points: 2 with section 9 ▼ · 4 with section 11 ▲

11

Section 11

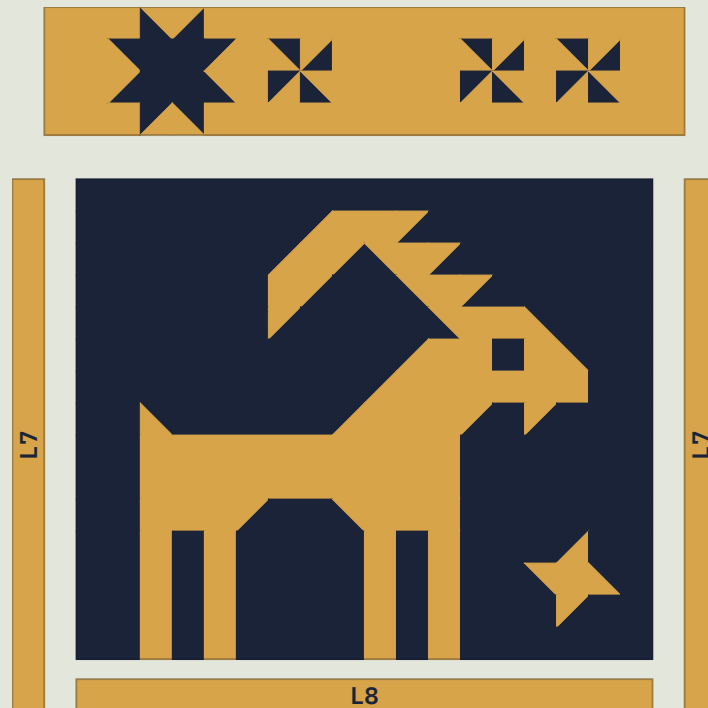
Four legs and the edelweiss.

Measures $36\frac{1}{2}'' \times 8\frac{1}{2}''$ Make unit **a** : join top to bottom **Edelweiss** **D3**Join left to right **D12** **L4** **D4** **L4** **D14** **L4** **D4** **L4** **D12** **a** **D4**

Pin points: 4 with section 10 ▼

STEP 5

Frame and assemble the top



Wide gaps are sewn last: window and L8 first, then the side strips, then the sky band.

1. Measure the window across its middle and trim **L8** to that length. Sew it to the bottom edge. The ibex now stands on solid ground.
2. Measure the height of window and L8 through the middle and trim both **L7** strips to that length. Sew them to the left and right edges.
3. Sew the sky band to the top. Match its ends with the side strips.
4. The quilt top measures $40\frac{1}{2}'' \times 40\frac{1}{2}''$. Stay-stitch around the edge, about $\frac{1}{8}''$ from the raw edge, to keep the outer seams from opening.

Quilt, bind and hang

Quilting

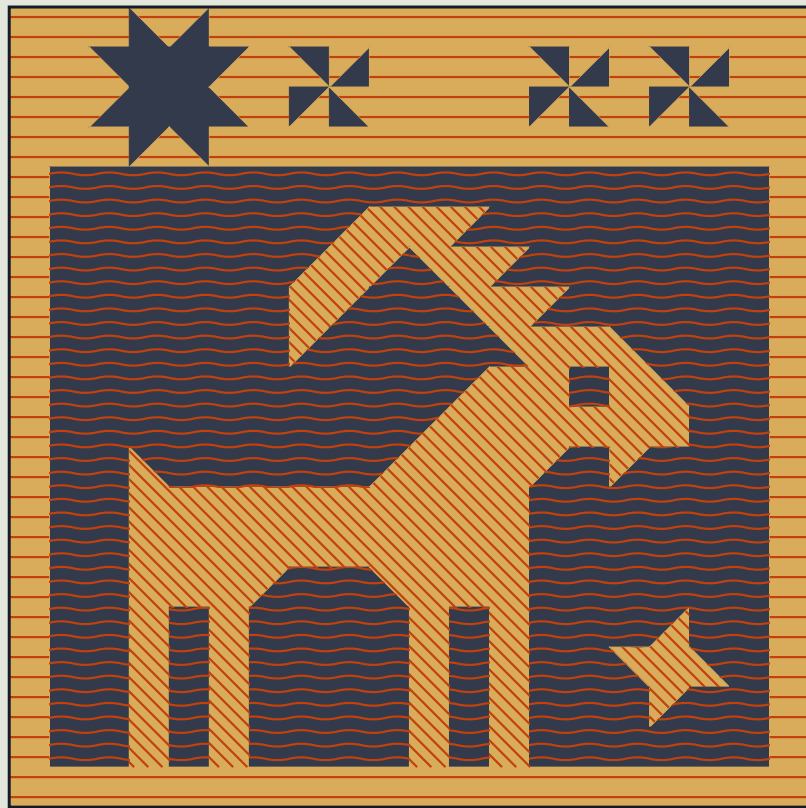
Layer the backing (right side down), the batting and the top. Baste with pins or spray. Then quilt:

1. Stitch in the ditch around the ibex, the stars and along the frame to stabilise the layers.
2. **Night:** gently wavy horizontal lines $\frac{5}{8}$ " apart, like wind over a ridge. A free-motion meander works too.
3. **Ibex:** straight diagonal lines $\frac{1}{2}$ " apart, parallel to the horn ridges. They read as fur.
4. **Band and frame:** straight horizontal lines 1" apart. Stop at the stars.

Use thread that blends into each fabric, or a slightly lighter gold on the night fabric for a starry shimmer.

Binding and hanging sleeve

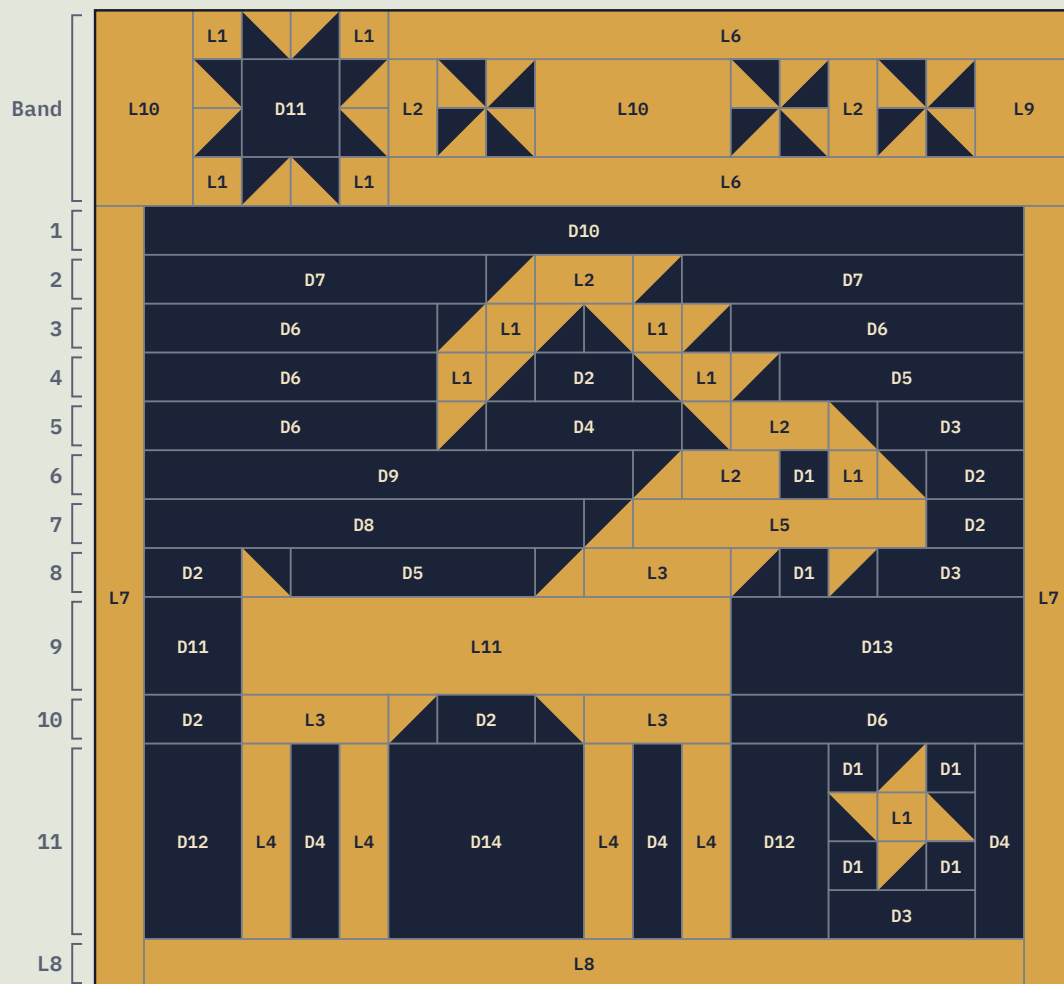
1. Trim the batting and backing even with the top.
2. Join the five binding strips end to end with diagonal seams and press them open. Fold the strip in half lengthwise, wrong sides together.
3. Hem the short ends of the sleeve strip. Fold it lengthwise, wrong sides together, and baste its raw edges to the top edge of the back. The binding seam will catch it.
4. Sew the binding to the front with a $\frac{1}{4}$ " seam, mitring the corners. Turn it to the back and hand-stitch it down.
5. Hand-stitch the folded lower edge of the sleeve to the backing. Leave a little slack so the rod does not distort the front.
6. Add a label with your name, the year and the star sign.



--- Suggested quilting lines

Layout chart

The whole quilt with every piece code. The brackets on the left mark the sky band, the eleven window sections and the bottom strip. Triangle units carry no code: there is only one kind.



Steinbock is an original design made as a companion to the patchwork lion zodiac quilt. Every measurement, cutting plan and diagram on this page is generated from one design grid and checked against it: each grid cell is covered exactly once, with the right fabric. The quilt was also sewn in software, seam by seam in the order given here, with real seam allowances and realistic sewing errors.

Star positions: Wolfram StarData. Names: δ Cap Deneb Algedi, γ Cap Nashira, β Cap Dabih, α Cap Algedi.